

Agent-based Intelligent Operations and Maintenance for Artificial Intelligence Data Centers

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Abstract—Driven by generative AI, the limitations of traditional manual models are highlighted by the severe operational challenges faced by Artificial Intelligence Data Centers (AIDCs), such as training hangs and performance bottlenecks. To address this, an innovative multi-agent operations and maintenance (O&M) framework has been designed and implemented. Within this framework, functions including intent recognition, task planning based on SOPs and model generation, tool execution incorporating log analysis and RAG, and end-to-end decision arbitration are seamlessly integrated through the modular collaboration of specialized agents (Coordinator, Planner, Tool, and Referee). Consequently, the end-to-end automation of fault diagnosis is achieved. The system's excellent performance has been validated on a benchmark dataset containing 80 injected faults, where a root cause localization accuracy of 85.16% was achieved. Furthermore, the core roles of the Standard Operating Procedure (SOP) knowledge base and the Referee agent in ensuring diagnostic efficacy and precision have also been confirmed through ablation studies. Ultimately, a validated technical path for building an efficient and intelligent O&M system for AIDCs is proposed, holding significant value for enhancing the stability and utilization of AI computing infrastructure.

Keywords—AIDC; Anomaly Detection; Artificial Intelligence for IT Operations; Large Language Model; Multi-Agent System; Root Cause Analysis; Fault Diagnosis

I. INTRODUCTION

In recent years, Generative AI technologies, represented by Large Language Models (LLMs), have achieved breakthrough progress, triggering a profound paradigm shift. The cornerstone of this shift is an unprecedented demand for massive computing resources, especially for compute infrastructure optimized for AI workloads. To meet this demand, the Artificial Intelligence Data Center (AIDC) has emerged. Unlike traditional data centers, an AIDC is a GPU-centric, heterogeneous computing cluster built around the characteristics of large-scale parallel computing. Its typical architectural features include tens of thousands of GPU compute nodes, low-latency lossless networks (like InfiniBand/RoCE) to support dense communication, and high-performance storage systems designed for massive parameters and datasets [1].

While providing powerful computing capabilities for various tasks, AIDCs, with their extreme performance and massive scale, also face unprecedented operational challenges. System complexity has caused failure modes to evolve from simple hardware failures to a series of subtle, intermittent, and cross-domain complex issues. In AI model training, especially

during distributed training, these problems manifest specifically as: the training task appears to run normally, but some nodes ("stragglers") exhibit low computation or communication efficiency, slowing down the overall training progress and causing GPU utilization to be far below its theoretical peak. The causes could involve any stage, such as data I/O, CPU preprocessing, or network congestion [2]. Furthermore, during distributed training, specific communication points (like All-Reduce) can stall completely without any error log output; these training hangs are often caused by hard-to-trace hardware sub-health, network deadlocks, or software defects. There are also unexpected crashes, typically caused by driver incompatibilities, library conflicts, ECC memory errors, or silent data corruption, which lead to training interruptions or NaN (Not-a-Number) results. The traditional O&M model cannot effectively address these complex problems as it overly relies on the experience of engineers and a reactive, "fire-fighting" mode driven by alerts and manual investigation [3]. In this model, personnel must switch between multiple independent tool stacks to complete the tedious process from anomaly detection in monitoring systems, to hardware diagnostics, application code bottleneck analysis, and log aggregation. Especially in a cluster environment with thousands of nodes, this traditional operational model is not only inefficient but also costly, severely impacting the effective utilization of computing power and the iteration efficiency of algorithms [4].

To solve the challenges faced by the traditional O&M model in AIDCs, academia and industry are exploring a more advanced operational paradigm: Agent-based Operations, also known as AIOps. Its core idea is to transform the manual troubleshooting process into an automated, proactive, and intelligent decision-making process with root cause analysis capabilities by building one or more intelligent agents [5]. The research in this paper aims to design an Agent-based O&M mechanism for AI training scenarios in AIDCs, with the goal of building a framework that can automatically diagnose typical performance bottlenecks and failures during training. Specifically, this research will systematically organize and define the key failure modes and their root-cause chains in AIDC training processes, and propose a lightweight, low-intrusion Agent workflow that can automatically integrate multi-dimensional data sources from monitoring, diagnostics, profiling, and logs [6]. Furthermore, it will explore and apply intelligent decision-making algorithms to automate the entire process from anomaly detection and fault correlation to root cause localization, significantly reducing the Mean Time To Resolution (MTTR) [7]. Through this research, we hope to

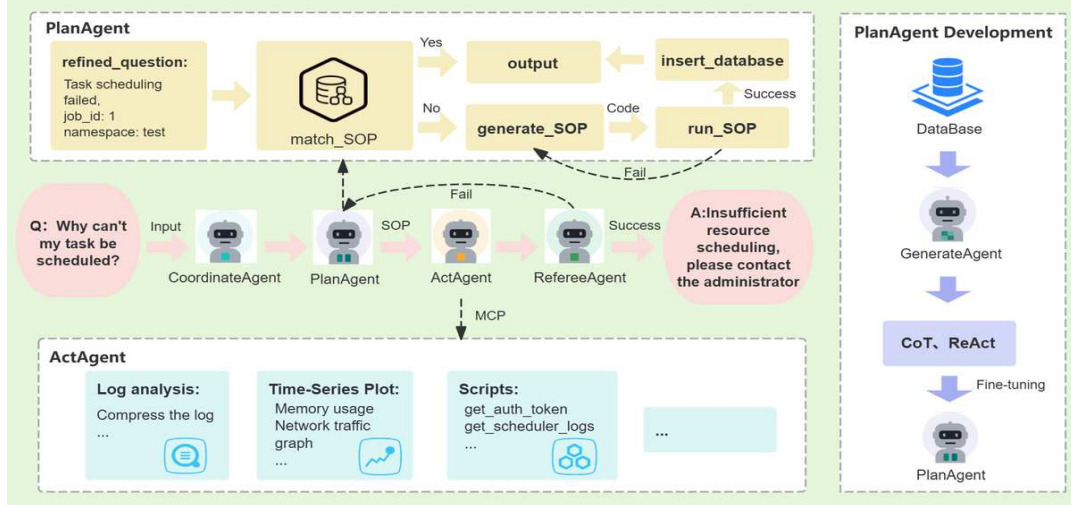


Fig. 1. Agent Process Flow

provide theoretical support and a feasible technical path for building an efficient and stable O&M system for AIDCs [8].

II. METHODOLOGY

A. Intent Recognition

The Coordinator Agent serves as the primary interaction entry point and intelligent distribution hub for the system. Its core task is to rapidly perform intent recognition and domain determination on the user's initial query. This agent acts as an efficient filter and sorter, ensuring that only valid requests related to the operations and maintenance (O&M) of the intelligent computing cluster can proceed to the subsequent complex processing pipeline. This lays the foundation for the entire system's precision and operational efficiency [9].

As shown in Fig. 1, the Coordinator Agent's working mechanism is based on a multi-class decision model. Upon receiving a user's Question, it immediately categorizes it into one of three predefined processing tracks. For non-task-oriented social interactions (e.g., greetings), the agent directly generates a friendly reply to maintain a positive user experience [10]. For requests that fall outside the scope of the O&M domain or are inappropriate, the agent performs a polite rejection, forming a security and efficiency barrier for the system. For all questions identified as valid O&M issues (such as those involving job scheduling, node status, or resource management), the agent triggers the core handoff operation.

Finally, by calling the `handoff_to_planner()` interface, the Coordinator Agent seamlessly transfers the identified valid O&M task to the downstream Planner Agent for in-depth analysis and resolution. This design separates simple interaction handling from complex task planning, creating a modular pipeline with clear responsibilities. This approach not only significantly reduces the computational overhead of the core functional modules through front-end filtering but also greatly enhances the system's response speed, robustness, and overall user experience [11].

B. Designated planning

A new paradigm in intelligent operations is being driven by the Planner Agent, a core component designed to expertly

orchestrate user requests and allocate tasks. Its primary function is to field incoming user requests and, through a sophisticated matching mechanism with a resource pool, determine if a pre-existing Standard Operating Procedure (SOP) is available for execution. Upon a successful match, the most suitable command is selected and dispatched to the next designated agent for implementation[12].

For requests that do not correspond to an existing SOP, the Planner Agent demonstrates its advanced capabilities by dynamically generating a solution. By leveraging fine-tuned small language models, the Planner Agent is equipped to autonomously create execution directives tailored to specific scenarios. This capability is underpinned by a deep understanding of the system's environment and an inference of the task's requirements, ensuring the generated instructions precisely meet the user's needs.

To guarantee the accuracy and stability of the entire process, the Planner Agent incorporates a "pre-execution validation" mechanism. This crucial step verifies the logic and executability of all instructions, whether they are matched from existing SOPs or newly generated. This pre-emptive validation allows for the early identification of potential errors, thereby preventing execution failures caused by improper commands and significantly enhancing the reliability and efficiency of the intelligent operations system.

Through this innovative design, the Planner Agent not only elevates the system's adaptive capabilities but also markedly improves the flexibility and intelligence of task processing, laying a robust foundation for the future of AIOps [13].

C. Monitoring Process

In the AIOps process, the RefereeAgent will identify and process the actions of the ActAgent. Specifically, after receiving an instruction from the ActAgent, the RefereeAgent will comprehensively analyze the information collected during the current Root Cause Analysis (RCA) process to evaluate whether the relevant root cause has already been identified. Based on this evaluation, the RefereeAgent will decide whether to approve (Yes) or reject (No) the execution according to predefined rules or strategies.

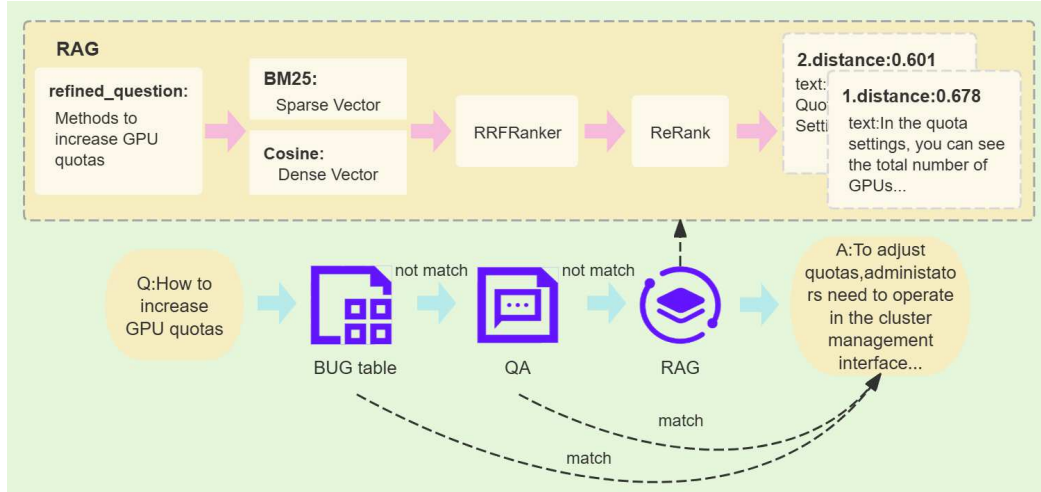


Fig. 2. Automatic Operating Process

Additionally, the RefereeAgent is also responsible for exception handling to ensure it can promptly discover and respond to potential risks during task execution. In particular, the RefereeAgent will conduct multi-dimensional monitoring and intervention based on the feedback returned by the tools used by the ActAgent. For example, when the Log analysis tool or a Time-Series Plot chart detects an anomaly (such as a memory overflow), the RefereeAgent will passively intervene, assess the severity of the anomaly, and decide whether to take responsive measures, such as terminating the task, to ensure system stability and safe task execution. Through this mechanism, the RefereeAgent not only plays a supervisory and dispatching role during task execution but also acts as a crucial component of the system's self-healing capabilities when exceptions occur, thus effectively enhancing the reliability and intelligence of the AIOps system [14].

D. Tools Agent

Within the AIOps framework, a Tool Agent boosts efficiency, accuracy, and intelligence by automating operational tasks. This simplifies workflows, minimizes manual effort, and ultimately leads to greater system stability.

Log parsing and analysis: In AIOps, network logs generate vast and redundant data. The Drain3 compression method efficiently and adaptively compresses these logs, significantly reducing storage requirements by eliminating redundancy while preserving crucial information. The compressed log data is then fed into an analysis module for efficient anomaly detection and performance analysis. This process enables the rapid identification of potential system issues and provides timely alerts for operations personnel, ultimately ensuring high system availability and stability [15].

Traffic prediction: In the tool agent architecture, key performance indicators such as second-level RDMA bandwidth, GPU-NIC throughput, and gradient synchronization time are continuously collected by a data acquisition component. The collected data is written to a time-series database and is normalized to ensure dimensional consistency for rapid modeling, thereby completely eliminating the need for manual intervention.

The agent's forecasting core is designed as a single-layer LSTM integrated with exogenous variable embeddings. Multi-step predictions are completed within millisecond-level inference latency, and the model's weights are periodically updated through rolling-window cross-validation to stably maintain prediction accuracy during volatile periods. As a result, mixed traffic patterns, combining both bursty and periodic characteristics, are accurately captured [16].

The generated forecast sequences are automatically written back to the scheduling layer. When low GPU utilization is predicted, actions such as job merging or migration are preemptively triggered. Through the autonomous control capabilities of this agent, inspection scripts are automatically invoked to diagnose the root causes of issues, whereby resource utilization and system stability are significantly improved.

Automated Operations: Upon receipt, the text of an external request is mapped into a unified embedding space, where its cosine similarity is calculated against the vectors of historical bug reports. If the highest similarity score reaches a 0.8 threshold, the request is automatically classified as a known defect. A standard reply, "The issue has been identified and is being fixed," is then triggered, and the subsequent computation chain is terminated. This mechanism ensures that confirmed defects are resolved via the shortest path, thereby avoiding the redundant consumption of computational and retrieval resources [17].

If a defect match does not occur, the request's embedding is reused to query the Question & Answer (QA) knowledge base. Should the similarity score again exceed the 0.8 threshold, the matched answer is delivered to the user. To enhance retrieval robustness, all new QA pairs are automatically expanded into three semantically equivalent questions upon ingestion, ensuring a high hit rate under diverse query formulations.

When neither of the first two stages yields an answer, the request is directed to a Retrieval-Augmented Generation (RAG) pipeline. Within this process, BM25 sparse retrieval and dense vector retrieval are first executed in parallel to construct a high-recall candidate set. Subsequently,

Reciprocal Rank Fusion (RRF) is employed to coarse-rank the dual results, followed by a fine-grained reranking performed by the bge-reranker-v2-m3 model. Finally, the top-k ranked documents are supplied as context to a generative decoder to formulate the final answer.

Through this three-tiered cascade of "Defect First, Knowledge Reuse, then Deep Retrieval," both high confidence and rapid response are simultaneously guaranteed. Comprehensive problem resolution and knowledge coverage in complex scenarios are achieved, significantly enhancing the automation and self-healing capabilities of the intelligent operations system.

Model Context Protocol(MCP): MCP, an open standard protocol, was introduced by Anthropic in November 2024 to unify the communication methods between LLMs and external data sources and tools. The potential of MCP is found in how the integration capability between AI models and operations-related systems can be enhanced by it, so that more efficient and intelligent operation and maintenance strategies can be realized [18].

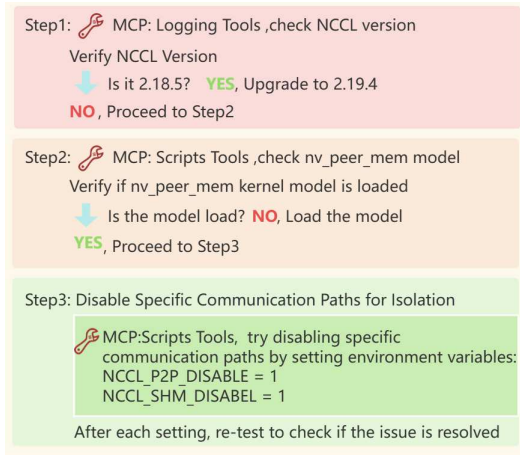


Fig. 3. MCP Processing Flow

When logs are observed to be persistently stuck in the "Init COMPLETE" stage by a log viewing tool called via MCP[1], the version of the NCCL communication library should first be verified. If version 2.18.5 is detected, a known GDR technical defect is considered to have been triggered, and it should be suggested to the customer that the version be upgraded to 2.19.4 or higher. Subsequently, the `nv_peer_mem` kernel module, used for peer-to-peer memory access between GPUs, should be checked via MCP. If the module has not yet been loaded, it should be loaded immediately and re-tested. If the module is ready but the stall persists, the issue can be isolated and diagnosed by specific communication paths being disabled until the normal initialization process is restored. For example, peer-to-peer communication can be disabled by setting `NCCL_P2P_DISABLE=1` via MCP, or shared memory communication can be disabled by setting `NCCL_SHM_DISABLE=1` [19].

III. DEMONSTRATION AND RESULTS

To validate the efficacy of our proposed model, the comprehensive evaluation metrics employed are outlined and a detailed analysis of the experimental results achieved is The testing and validation of the overall experiment are primarily

divided into two parts: validation of the overall workflow and validation of the individual tool components.

A. Overall Metrics

For the empirical evaluation of the multi-agent system proposed in this research, a series of rigorous validation experiments were designed and executed in a high-performance intelligent computing cluster environment. To ensure the comprehensiveness and authenticity of the evaluation, a benchmark dataset comprising 80 independent anomalous events was constructed. The generation of this dataset was achieved through the Systematic Fault Injection of eight predefined fault types, which were designed to precisely simulate common challenges in real-world operations, such as NVIDIA NCCL version conflicts and GPU computing resource preemption scenarios.

The Root Cause Analysis (RCA) efficacy of the system was quantitatively evaluated using two Key Performance Indicators (KPIs): Location Accuracy (LA) and Type Accuracy (TA). The calculation of these two metrics was strictly formalized as follows:

$$LA = \frac{L_c - \sigma \cdot L_i}{L_t}; TA = \frac{T_c - \sigma \cdot T_i}{T_t}$$

In the formulas above, L_c and T_c are defined as the counts of correctly identified root cause locations and types, respectively, whereas L_i and T_i are the corresponding counts of incorrect identifications. L_t and T_t represent the total number of ground truth instances. It is noteworthy that a penalty factor, sigma, is introduced into the formulas. Its purpose is to impose a more significant negative weight on incorrect identifications, thereby enabling a stricter measurement of the model's precision.



Fig. 4. Results of the Ablation Study

As shown in Fig. 4, to conduct an in-depth analysis of the functional contributions of the system's internal components, a comprehensive Ablation Study was also specifically carried out. The core methodology of this study was established as systematically and temporarily removing a single agent and re-evaluating performance under this "degraded" configuration. The degree of performance degradation in LA and TA metrics, caused by the absence of a specific agent, was to be precisely quantified. Based on this quantification, the importance and indispensability of each independent agent within the entire collaborative analysis framework could be objectively assessed.

The ablation experimental results are presented in Figure 2. According to the experimental results, the SOP knowledge

base was identified as the cornerstone of the system's capabilities, and its integrity was confirmed to be the key determinant of the lower bound of diagnostic efficacy. Concurrently, the RefereeAgent was highlighted as a pivotal role in ensuring diagnostic precision, with its arbitration of the final decision being regarded as an essential step for achieving high accuracy. The role of the CoordinateAgent was positioned as a facilitator of process optimization, ensuring efficient coordination within the system. Therefore, the exceptional RCA performance of the system can be attributed to the indivisible synergy among core knowledge, decision arbitration, and process coordination, wherein the unique function of each component was proven to be necessary for achieving overall optimal performance.

B. Tools Metrics

Each tool was subjected to dedicated testing, and both testing and validation were carried out using conventional verification methods.

Log parsing and analysis: For this validation and test, 72 hours of continuously generated operational logs were collected from the Spine-Leaf switching network and GPU servers of the intelligent computing center. Three types of parsers (Drain3, Spell, and LLM) were deployed separately, and their performance was evaluated based on metrics including Precision, Recall, Exact Match Ratio (representing the proportion of high-confidence template matches), and average parsing latency.

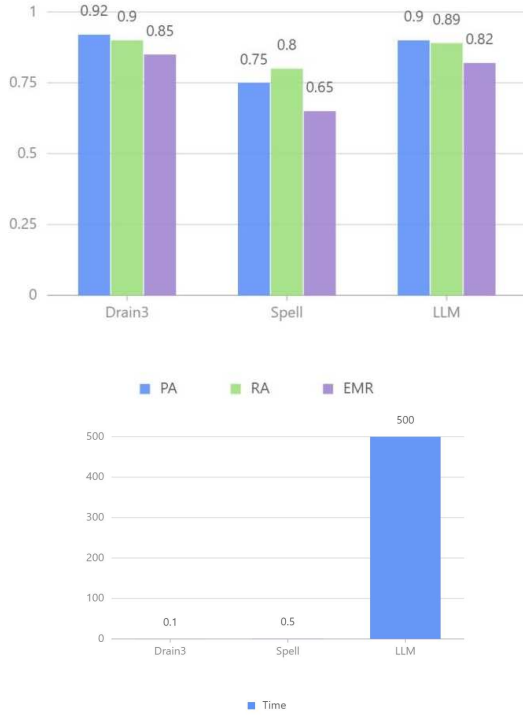


Fig. 5. Results of the Log Parsing

As shown in Fig. 5, the applicability of the Drain3 fixed-depth prefix tree strategy for handling high-concurrency, long-log scenarios was verified. For the parsed logs, further fault localization and root cause analysis can be performed. Consequently, the efficiency of the Tool Agent in fault

localization and resource scheduling within intelligent AIOps scenarios is significantly improved.

Traffic prediction: To validate the model's performance, an experimental dataset sourced from a 72-hour LLMs training process was utilized. This dataset comprises second-level monitoring time series, including key metrics such as RDMA bandwidth, GPU-NIC throughput, and gradient synchronization time.

Predictive performance was benchmarked against two baselines: SARIMA, representing classical statistical approaches, and GRU, a lightweight recurrent neural alternative. Three complementary error metrics were adopted to quantify absolute deviation, prediction stability, and relative accuracy between forecasts and ground truth.

Mean Absolute Error (MAE):

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Root Mean Squared Error (RMSE):

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

Mean Absolute Percentage Error (MAPE):

$$MAPE = \frac{100\%}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

As shown in Fig. 6, the experimental results indicate that the proposed LSTM-based model significantly outperforms baseline models such as SARIMA and Prophet across all evaluation metrics. Its fundamental advantage lies in its ability to effectively learn complex non-linear dependencies within the data and accurately capture the complex "bursty-periodic" mixed patterns characteristic of LLM training traffic.

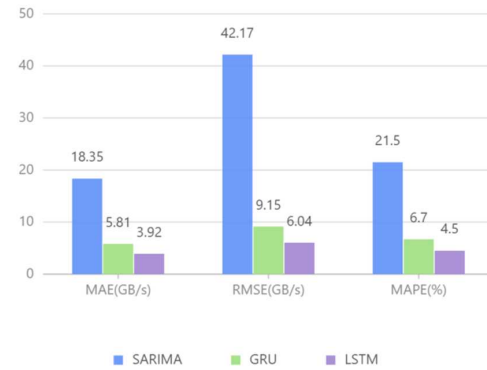


Fig. 6. Results of the Traffic Prediction

This demonstrates that the model possesses greater sophistication and practical value in handling such dynamic, non-stationary time series data compared to traditional models that rely on fixed periodicity assumptions.

Automatic operation: To quantitatively evaluate the effectiveness of the "defect priority—knowledge reuse—deep retrieval" three-stage intelligent operation framework, a rigorous verification process was constructed. This process is

designed to test the end-to-end performance of the system, from problem understanding to answer generation, in a real-world operation and maintenance scenario. The industry-recognized RAG system evaluation framework, RAGAS, was adopted, and 49 test questions were automatically generated based on the operation and maintenance knowledge base to comprehensively simulate user requests.

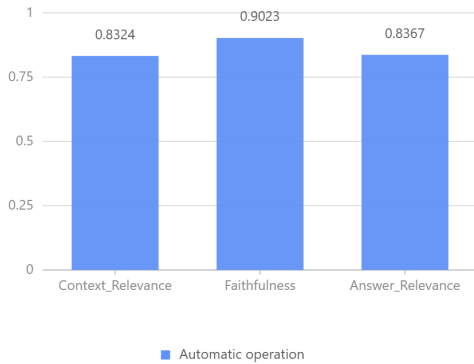


Fig. 7. Results of the Automatic Operation

As shown in Fig. 7, the experimental data strongly demonstrate the superior performance of this three-stage automated operation framework. As shown in the table above, the system performs excellently on all indicators. A faithfulness score of up to 0.9023 was achieved, highlighting the system's high degree of respect for factual evidence when generating answers, effectively avoiding the production of misleading information. At the same time, both the context relevance and answer relevance scores exceeded 0.83, indicating that the system can not only accurately retrieve key information from massive documents but also construct concise and effective answers that are highly consistent with user intent based on this information. In summary, this quantitative evaluation has verified the core value of the framework in improving the level of automation in operations and maintenance, and in guaranteeing response speed and accuracy.

IV. CONCLUSION AND FUTURE WORK

The limitations of the traditional O&M model have been demonstrated, and the urgent need to build an automated, intelligent O&M system has thereby been highlighted. To this end, a new O&M paradigm based on intelligent agents has been proposed and designed. Through this paradigm, the fault resolution process is intended to be transformed from a reactive, manual "fire-fighting" approach dependent on human experience into a proactive process equipped with predictive analysis and automated decision-making capabilities.

Through the Agent O&M framework designed in this research, a lightweight, low-intrusion workflow has been constructed, by which full-process automation—from anomaly detection to root cause localization—is achieved. Via this framework, the Mean Time To Resolution (MTTR) is expected to be significantly shortened, and the utilization efficiency of computing resources is expected to be maximized. More importantly, for the construction of next-generation, high-efficiency, and stable AIDC O&M systems,

key theoretical support and a feasible technical path have been provided. This is of significant importance for promoting the continuous development of generative AI technologies.

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